Indian Institute of Science Education and Research, Kolkata
Math Methods (PH2204)
Midsem Exam, Date: 7 March, 2025

Timing: 15:00 to 16:30 | Spring 2025 | Max mark: 100

Solve all questions, each carry [20] marks.

Notation: $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \equiv (x, y, z) \equiv (r\sin\theta\cos\phi, r\sin\theta\sin\phi, r\cos\theta)$; $\nabla \equiv (\partial_x, \partial_y, \partial_z) \equiv (\partial_r, \frac{1}{r}\partial_\theta, \frac{1}{r\sin\theta}\partial_\phi)$.
$A = |\vec{A}| = \sqrt{\vec{A}.\vec{A}}$ $\quad x \equiv x(t)$, $\dot{x} = \frac{d}{dt}x(t) \equiv Dx(t)$, $\ddot{x} = \frac{d^2}{dt^2}x(t) \equiv D^2x(t)$, $i = \sqrt{-1}$

1. **(a)** Express ∇ in (r, θ, ϕ)-coordinate system, where $x = r\cos\theta\sec\phi, y = r\sin\theta\sec\phi, z = r\tan\phi$. **[10]**
   (b) Use Levi Civita symbols to prove $\vec{A} \times (\nabla \times \vec{A}) = \frac{1}{2}\nabla A^2 - (\vec{A}.\nabla)\vec{A}$ **[10]**

2. **(a)** Find an analytic function $f(z)$ whose real part is $x^2 + xy - y^2$. **[10]**
   (b) Find regions in complex plane where both $f_1(z) = |y| - ix$ and $f_2(z) = |x| + i|y|$ are analytic. **[10]**

3. Find the unit vectors which are parallel to the tangent line of the curves $\Re(ze^z)$ and $\Im(ze^z)$ at $z = z_0 = 1 + i$, and show that they are orthogonal. Use Cauchy–Riemann criteria to show that $z_0 = 1 + i$ is not special, the orthogonality holds at any other arbitrary point $z_0 = a + ib$. **[20]**

4. **(a)** Use variable transformation to convert the equations $\{\dot{x} = x\ln y; \dot{y} = -y\ln x\}$ to linear form and solve them using matrix method for initial condition $x(0) = e^2$ and $y(0) = 1/e$ **[10]**
   (b) Show that particular integral of the diff eqn $(D - a)x(t) = f(t)$ is $e^{at}\int_0^t f(t')e^{-at'}dt'$. **[10]**

5. **(a)** Given a matrix $M = \frac{1}{5}\begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$, compute M^n **[12]** **(b)** What is $\lim_{n\to\infty} M^n$? **[8]**